

Mehar Klair

587-568-5589 | meharsinghklair@gmail.com | linkedin.com/in/mehar-klair | github.com/meharklair | [portfolio website](#)

EDUCATION

University of Alberta

Bachelor of Science in Computer Science, Minor in Business

Edmonton, AB

Sept. 2022 – Apr 2027

EXPERIENCE

Machine Learning Research Assistant

May 2025 – Present

Data and Network Security Lab, University of Alberta

Edmonton, AB

- Engineered over **60,000** complex data samples into **LLM** friendly graph representations using **Python**, ensuring scalability and increasing model processing efficiency by **30%**.
- Reduced data preprocessing time by **40%** by implementing data pipelines that transformed raw system logs into structured graph data.
- Developed **LLM** prompts using industry AI prompting techniques to generate clear attack summaries from data graphs, improving report accuracy by **20%**.
- Translated cutting-edge research into practical solutions, applying academic techniques to optimize data/prompt strategies.
- Collaborated with a small research team using Agile practices to align technical approaches with project objectives, ensuring high-quality and impactful research outcomes.

TECHNICAL PROJECTS

Threat scanner | *Python, Yara, Urllib3* | [GitHub](#)

Aug 2024 – Present

- Developed a threat scanning tool in **Python** using Yara rules and a threat signature database to detect malicious files.
- Utilized **API** calls to retrieve detailed information on malicious files, reducing manual analysis time by **21%** and enabling quick remediation.
- Increased threat coverage by **45%** by automating the retrieval of over **800,000** threat signatures.
- Enhanced **scalability** by developing a **modular system** for plug-and-play integration of future threat intelligence sources.

Brazmas | *Android Studios, Java, Figma, Firebase, UML* | [GitHub](#)

Jan 2024 – Apr 2024

- Developed an event check-in Android application in **Java** enabling real-time attendance and push notifications.
- Reduced redundant code by **31%** and improved maintainability, by designing an **API** for communication between the Android client and the **Firebase** backend.
- Achieved **93%** test coverage by writing 60 unit tests with **JUnit** and 25 integration tests with **Espresso**, improving app stability by **28%**.
- Optimized **database queries** and **Firebase** interactions, reducing response times by **40%**, and enhancing user experience during high-traffic events.

EXTRA-CURRICULAR EXPERIENCE

Lead CTF Challenge Developer

Oct 2024 – Present

Cybersecurity, Hacking and Digital Security Club

Edmonton, AB

- Designed a **Python bytecode reverse-engineering** challenge. Forcing participants to reconstruct logic from **.pyc** files to uncover a **code injection vulnerability**, solved by **36%** of teams.
- Directed event planning and coordination for a Capture the Flag event, resulting in a **50%** increase in sign-ups compared to the previous year.
- Increased average solve rate by **42%** by leading the development of 10 challenges across 5 categories, focusing on cohesive and engaging designs to enhance participant experience.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, C, C#, HTML, CSS, RISC-V, SQL

Frameworks/Tools: Git, Pandas, Figma, MongoDB, PyTorch, Scikit, TensorFlow, Firebase, Junit, GitHub, Linux, Nmap, Espresso, Wireshark

Relevant Coursework: Operating Systems, Databases, Machine Learning, Object-Oriented Programming, Data Structures and Algorithms, and Software Engineering.